Econ 330 Essay:

Exploring the UK Real Estate Prices: A Econometric Analysis of Economic Indicators

Motivation:

The UK real estate market is of great importance to economists, policymakers and consumers, and harnesses a wide array of economic interactions due the how dynamic the market is. The relevant figures of the real estate price can be interpreted and translated into economic health of a country, but it can also have a direct impact on consumer wealth and how consumers behave whilst spending. This means that real estate price index captures the influences it has on microeconomic decisions and macroeconomic policies.

Research has consistently portrayed that real estate prices are intricately linked to a host of economic indicators such as interest rates, GDP growth and inflation. Ahearne et al. (2005) demonstrates strong correlation between interest rates and housing price, highlighting that when the interest rates are lower, it causes house prices to exist since it lowers the mortgage borrowing costs, which will push the price upwards regardless. This is like GDP growth, when there is growth, it tends to increase the economic activity and consumer confidence, which results in a boost in demands for housing (Case & Shiller, 2003).

The impact of macroeconomic stability on real estate prices cannot be overstated. It is common knowledge to think that during economic downturns and recessions, it typically leads to a decline in property values, which was observed during the global financial which resulted in the Uk housing price fall sharply (Miles % Pillonca, 2008).

Inflation also has 2 effects on real estate prices. During times of moderate inflation, it may encourage real estate as a hedge against the devaluing of money, however high inflation experiences adverse effects, making housing less affordable therefore decreasing in demand. (Anari & Kolari, 2002). With regards to consumer confidence, as a barometer of economic sentiment, significantly influences housing market dynamics. An optimistic outlook on the economy can lead to increased buying activity, while pessimism due to economic uncertainty can dampen the market (Siskos et al., 2013).

Given these factors, this essay dives into exploring the relationships between the UK’s real estate price index and various macroeconomic indicators, using a comprehensive econometric framework. Using these techniques, the aim is to uncover the underlying mechanisms driving the UK housing market.

Summary of the findings as follows:

- Using an instrumental variable GBPtoEUR to instrument interest rates to check for any endogeneity.
- Time series regression analysis

Econometric model:

RealEstatePrice = B0 + B1InterestRate + B2GDP + B3UnemploymentRate + B4InflationRate + B5ConsumerConfidence + B6Recession + Error Term

This is a multiple linear regression model. It would be estimated using OLS methods, including robust standard error accounting for heteroscedasticity. There is potential endogeneity of interest rates, which would address using an instrumental variable GBP to EUR exchange rates using Two-Stage Least Squares (2SLS) regression. Also, I used time series regression such as autoregression and distributive lag.

Key Variables:

This dataset consists of monthly panel data during 1985 – 2020 focusing on the UK. Due to the lack of data within the desirable time frame for the Real Estate Price and GPD variable, I had managed to find quarterly data dating from that period, where I would the use the interpolate command function in Stata to fill in the data as monthly. My findings have come from a variety of sources such as the ONS, Bank of England.

Interest Rates: Interest rates are a critical determinant of mortgage rates, and hence influencing housing affordability and demand. To simplify this into a monthly dataset, I took the latest interest rate change for that specific month if that month had experienced multiple changes in interest rates.

GDP growth rate: This was a quarterly growth rate the UK's Gross Domestic Product, which is adjusted for inflation. It is a primary indicator of economic health and is expected to positively affect real estate prices due to its relationship with income and employment levels.

Inflation Rate: This measurement is the year-on-year percentage change in the Consumer Price Index. Inflation affects the real value of money, resulting in influencing the real estate investment decisions.

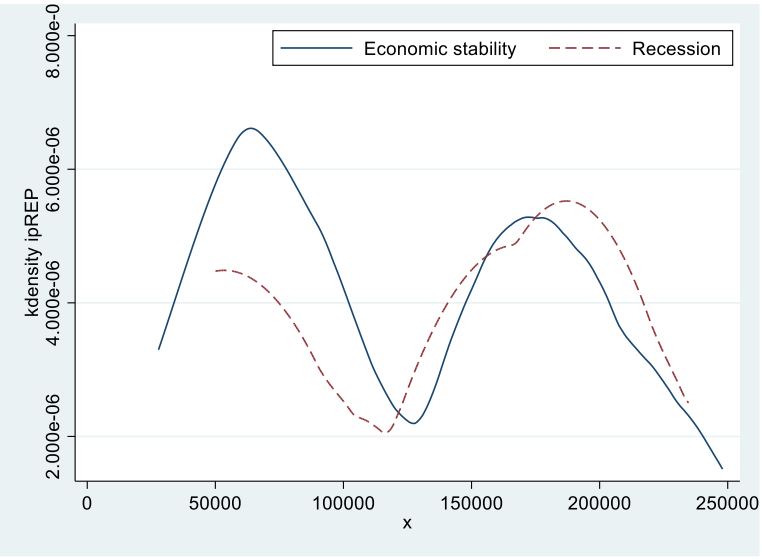
Unemployment: Many people who are unemployed for long periods of time result to claiming benefits or live in a council estate. This can be an indicator that the overall UK public are not earning a minimum wage, which makes it difficult for earning a mortgage, or being able to consider buying a house with financial stability

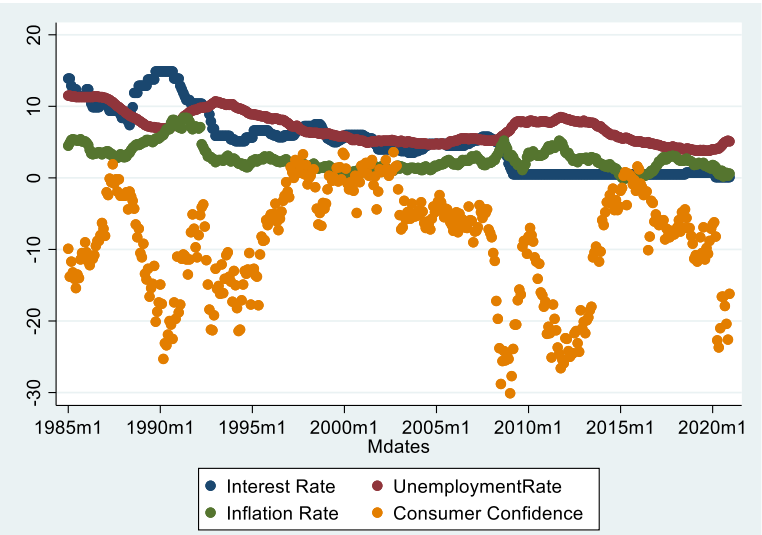
Consumer Confidence: This is an index of the overall public's consumer sentiment towards the economy, which is derived from a survey data. If it more important to

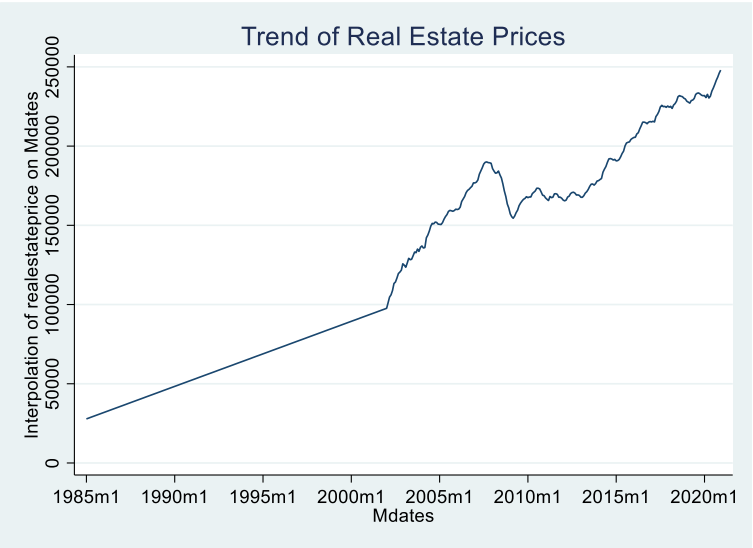
save and cut costs rather than splash out more money on more luxurious items, it could be a direct effect on individuals spending behaviour when buying a house.

Recession: this reflects the UK economy experiencing 2 consecutive quarters of the economy having a negative growth or shrinking due to less productivity. The purpose of this dummy variable is to capture the cyclical downturns in the economy that potentially depress real estate prices.

GBPtoEUR (instrumental variable): This is used to instrument interest rates in this study. The exchange rate is influenced by similar macroeconomic factors that affect interest rates. However, we believe that it is exogenous to the real estate price in the UK. This helps to address any potential endogeneity between interest rates and real estate prices, eliminating biased estimations and ensures reliability.


Economic stability
Recession
kdensity ipREP
8.000e-0
6.000e-06
4.000e-06
2.000e-06
0
50000
100000
150000
200000
250000
x


20
10
0
-10
-20
-30
1985m1
1990m1
1995m1
2000m1
2005m1
2010m1
2015m1
2020m1
Mdates
Interest Rate
UnemploymentRate
Inflation Rate
Consumer Confidence


Trend of Real Estate Prices
Interpolation of realestateprice on Mdates
0
50000
100000
150000
200000
250000
1985m1
1990m1
1995m1
2000m1
2005m1
2010m1
2015m1
2020m1
Mdates

Regression Analysis:

OLS, Robust OLS and 2SLS regression.

```
. reg ipREP interestrate ipGDP unemploymentrate inflationrate consumerconfidence recession
```

Source	SS	df	MS	Number of obs	=	432
				F(6, 425)	=	2415.40
Model	1.7860e+12	6	2.9766e+11	Prob > F	=	0.0000
Residual	5.2375e+10	425	123234190	R-squared	=	0.9715
				Adj R-squared	=	0.9711
Total	1.8383e+12	431	4.2653e+09	Root MSE	=	11101

ipREP	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
interestrate	3533.244	490.9636	7.20	0.000	2568.225	4498.264
ipGDP	.6223753	.0184098	33.81	0.000	.5861898	.6585608
unemploymentrate	793.5056	565.3017	1.40	0.161	-317.6297	1904.641
inflationrate	152.332	582.9087	0.26	0.794	-993.4109	1298.075
consumerconfidence	-23.66757	101.5756	-0.23	0.816	-223.3206	175.9855
recession	-5390.072	1968.348	-2.74	0.006	-9258.981	-1521.163
_cons	-96911.9	10602.55	-9.14	0.000	-117751.9	-76071.94

```
. reg ipREP interestrate ipGDP unemploymentrate inflationrate consumerconfidence recession, robust
```

Linear regression

Number of obs	=	432
F(6, 425)	=	4233.21
Prob > F	=	0.0000
R-squared	=	0.9715
Root MSE	=	11101

ipREP	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
interestrate	3533.244	466.7067	7.57	0.000	2615.904	4450.585
ipGDP	.6223753	.0171825	36.22	0.000	.5886021	.6561486
unemploymentrate	793.5056	524.2702	1.51	0.131	-236.9798	1823.991
inflationrate	152.332	429.5893	0.35	0.723	-692.0522	996.7161
consumerconfidence	-23.66757	82.50693	-0.29	0.774	-185.84	138.5049
recession	-5390.072	1597.392	-3.37	0.001	-8529.844	-2250.299
_cons	-96911.9	9944.145	-9.75	0.000	-116457.7	-77366.07

```
. ivreg ipREP ipGDP unemploymentrate inflationrate consumerconfidence recession (interestrate=gbptoeur)
```

Instrumental variables 2SLS regression

Source	SS	df	MS	Number of obs	=	432
				F(6, 425)	=	2232.84
Model	1.7819e+12	6	2.9698e+11	Prob > F	=	0.0000
Residual	5.6469e+10	425	132867697	R-squared	=	0.9693
				Adj R-squared	=	0.9688
Total	1.8383e+12	431	4.2653e+09	Root MSE	=	11527

ipREP	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
interestrate	6363.14	3423.645	1.86	0.064	-366.2449	13092.52
ipGDP	.719235	.117442	6.12	0.000	.4883956	.9500744
unemploymentrate	2944.078	2638.896	1.12	0.265	-2242.834	8130.99
inflationrate	-1894.872	2522.806	-0.75	0.453	-6853.603	3063.858
consumerconfidence	22.52846	119.0732	0.19	0.850	-211.5172	256.5741
recession	-5525.723	2050.268	-2.70	0.007	-9555.651	-1495.795
_cons	-151205.6	65879.38	-2.30	0.022	-280695.6	-21715.65

This econometric analysis regressed using stata shows: Ordinary Least Squares (OLS), OLS with robust standard errors and Two-stage Least Squares (2SLS) this incorporates an instrumental variable.

The basic OLS regression has shown an interesting relationship between the economic indicators. It consists of a significant positive correlation between interest

rates and property prices. However, this is counterintuitive as it suggests that the higher the interest rates, the higher the borrowing costs whilst suppressing property prices, which would discourage buy houses. This may coincide with periods of economic optimism or other market dynamics not captured explicitly in this model. In addition, the OLS shows that GDP has a positive effect, confirming the hypothesis that economic growth leads to higher property prices and economic prosperity.

In contrast unemployment rate did show a positive relationship, yet did not significantly impact property prices, which may suggest that its effect might be overshadowed by other stronger market forces or economic indicators. This is the same for inflations too. For consumer confidence, theoretically we would have expected it to be positively correlated with real estate price, yet shows a negative but non-significant relationship, indicating consumer sentiment might not be a direct driver of real estate prices with the current model. However, the regression shows that during a recession it does have a negative effect on property prices, resulting that there is sensitivity of the real estate market to recession.

When accounting for potential heteroscedasticity, we use robust standard errors, which causes the significance and direction of the coefficients remain largely consistent with the basic OLS model. This reassures the reliability of the initial findings. The robustness check confirms the influential roles of the interest rates and GDP, as well as the recession's impact, indicating that these findings are not artifacts of specific data issues such as heteroscedasticity.

The use of 2SLS regression is to address any endogeneity issues that could potentially be present, which in our case we instrument interest rate with GBP to EUR exchange rate, which shows there are some notable divergences from the OLS results. The coefficient of interest rates in the 2SLS model shows that is greater than the compared to both the normal OLS and robust OLS, suggesting that once endogeneity is controlled for, the effects of interest rates on real estate prices might be more substantial that previously estimated. This means that our instrument variable underscores the importance of addressing endogeneity to uncover the true economic relationship.

Autoregressive models:

Time Series Regression:

```
. reg ipREP Mdates
```

Source	SS	df	MS	Number of obs	=	432
				F(1, 430)	=	11598.62
Model	1.7726e+12	1	1.7726e+12	Prob > F	=	0.0000
Residual	6.5717e+10	430	152830088	R-squared	=	0.9643
				Adj R-squared	=	0.9642
Total	1.8383e+12	431	4.2653e+09	Root MSE	=	12362

ipREP	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
Mdates	513.6577	4.769475	107.70	0.000	504.2833	523.0321
_cons	-139976.9	2529.586	-55.34	0.000	-144948.8	-135005

The purpose of starting off with reviewing this regression model is to give an insight of the overarching trend real estate prices has over the period we are studying. We regress real estate price against time, which shows us what direction the pace of the market changes. The regression shows us that the coefficient is positive and is significant associated with the time variable Mdates, resulting in a steady increase in property price, which could be attributed to a long-term economic growth. This upward trend provides a guideline understanding of the market's response to ongoing economic conditions.

Distributed Lag Model:

```
. reg ipREP interestrate_1 ipGDP_1 unemployment_1 inflationrate_1 consumerconfidence_1 recession_1
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Source	SS	df	MS	Number of obs	=	431
				F(6, 424)	=	2389.36
Model	1.7764e+12	6	2.9606e+11	Prob > F	=	0.0000
Residual	5.2537e+10	424	123908195	R-squared	=	0.9713
				Adj R-squared	=	0.9709
Total	1.8289e+12	430	4.2533e+09	Root MSE	=	11131

ipREP	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
interestrate_1	3543.214	495.3698	7.15	0.000	2569.528	4516.9
ipGDP_1	.6232757	.0185758	33.55	0.000	.5867635	.6597878
unemployment_1	806.9764	568.2909	1.42	0.156	-310.0417	1923.995
inflationrate_1	96.42521	587.5524	0.16	0.870	-1058.453	1251.303
consumerconfidence_1	6.733698	102.0043	0.07	0.947	-193.7633	207.2307
recession_1	-5774.678	1974.714	-2.92	0.004	-9656.126	-1893.23
_cons	-96383.35	10685.91	-9.02	0.000	-117387.3	-75379.4

This regression offers us a nuanced exploration of how various economic indicators influence real estate prices both immediately and over time. We regressed all the independent variables that was listed at the beginning, which then resulted in measuring their direct and lagged impact on the housing market. Results indicate that GDP growth is significant and had a positive effect, which confirms that economic propensity increases market expansion. Alternatively, interest rates exhibit

a complex relationship since their impact became significant after certain lags. This can reflect the time it takes for monetary policy adjustments to take affect into the economy which has a knock-on effect to consumer behaviour and investment decisions.

Autoregressive model:

```
. reg ipREP ipREP_1 ipREP_2 ipREP_3 ipREP_4 ipREP_5 ipREP_6 ipREP_7 ipREP_8 ipREP_9 ipREP_10 ipREP_11 ipREP_12
```

Source	SS	df	MS	Number of obs	=	420
				F(12, 407)	>	99999.00
Model	1.7263e+12	12	1.4386e+11	Prob > F	=	0.0000
Residual	374218699	407	919456.262	R-squared	=	0.9998
				Adj R-squared	=	0.9998
Total	1.7267e+12	419	4.1209e+09	Root MSE	=	958.88

ipREP	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
ipREP_1	1.324645	.0495566	26.73	0.000	1.227226	1.422064
ipREP_2	-.1343702	.081997	-1.64	0.102	-.2955607	.0268204
ipREP_3	.0721714	.0822865	0.88	0.381	-.0895882	.2339311
ipREP_4	-.1433347	.0822928	-1.74	0.082	-.3051067	.0184373
ipREP_5	-.3647908	.0822344	-4.44	0.000	-.526448	-.2031336
ipREP_6	.0853564	.0841322	1.01	0.311	-.0800315	.2507444
ipREP_7	.0688679	.0843619	0.82	0.415	-.0969716	.2347074
ipREP_8	.179454	.0823276	2.18	0.030	.0176137	.3412944
ipREP_9	.0693003	.0834707	0.83	0.407	-.0947873	.2333878
ipREP_10	-.0645508	.0852235	-0.76	0.449	-.2320839	.1029823
ipREP_11	-.1360587	.0858391	-1.59	0.114	-.304802	.0326846
ipREP_12	.0435567	.0515416	0.85	0.399	-.0577642	.1448776
_cons	132.7811	106.1612	1.25	0.212	-75.91155	341.4738

I used this to test to find out whether previous real estate prices influence future prices further exploring the market dynamics. This model shows the past price levels are highly predictive of future trends, suggesting a strong market confidence and a tendency for price persistence. Due to the significance of the coefficients on lagged price terms, it illustrates the cyclicality within the real estate market, potentially driven by speculative behaviours or investor expectations based on past performance.

Dickey-Fuller Test for stationarity:

```
. dfuller ipREP

Dickey-Fuller test for unit root                   Number of obs   = 431
Variable: ipREP                                    Number of lags  =   0

H0: Random walk without drift, d = 0

                     Dickey-Fuller
                    critical value
                Test ---------------------------------
            statistic          1%          5%         10%
------------------------------------------------------------
 Z(t)           1.620      -3.446      -2.873      -2.570
------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.9979.
```

All though the regressions above may seem credible to begin with, we reliability can check its reliability using a dickey-fuller test to check for non-stationarity is the data set, potentially causes misleading results. The score indicates that it is non-

stationary, which implies that there are trends with varying variances over time. This is a critical finding, which justifies the necessity for models capable of handling such data characteristics, ensuring that the analysis does not lead to spurious conclusions.

Arima Model:

ARIMA regression

Sample: 1985m1 thru 2020m12 — Number of obs = 432
Wald chi2(12) = 10182.87
Log likelihood = 1271.673 — Prob > chi2 = 0.0000

ln_ipREP	Coefficient	OPG std. err.	z	P>|z|	[95% conf.	interval]
ln_ipREP						
_cons	11.55946	.0446	259.18	0.000	11.47205	11.64687
ARMA						
ma						
L1.	2.96227	.0527338	56.17	0.000	2.858914	3.065626
L2.	5.960724	.1674118	35.61	0.000	5.632603	6.288845
L3.	9.151784	.2835263	32.28	0.000	8.596082	9.707485
L4.	12.50378	.4273762	29.26	0.000	11.66614	13.34142
L5.	14.98152	.5672067	26.41	0.000	13.86982	16.09323
L6.	15.65326	.6429842	24.34	0.000	14.39304	16.91349
L7.	14.60419	.6389084	22.86	0.000	13.35195	15.85642
L8.	12.19686	.5664241	21.53	0.000	11.08669	13.30703
L9.	8.831014	.4417525	19.99	0.000	7.965195	9.696833
L10.	5.545206	.2937851	18.88	0.000	4.969397	6.121014
L11.	2.788597	.1641948	16.98	0.000	2.466781	3.110413
L12.	.8985966	.0731887	12.28	0.000	.7551495	1.042044
/sigma	.0093157	.0005168	18.03	0.000	.0083027	.0103286

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

This is used to eliminate the effects of potential spurious regression from non-stationary data due to it integration of autoregressive and moving average components. In also incorporated a log transformation to the real estate price to further eliminate the effects of non-stationary data. This enhances the effect of the ARIMA model, allowing for more detailed examination of how short/long term shocks and longer-term trends impact real estate prices. The moving average terms reveal significant insights into the effects of economic conditions over each successive periods, highlighting the enduring impact of past economic shocks on current and future market conditions.

Conclusion:

The combination of 2SLS and Time Series econometric analysis has allowed us to understand the dynamics that affect the UK real estate prices. Despite the weakness of the instrumental variable, it can be said that for this model, there is not a close match for an instrumental variable when regressing against real estate price.

In addition, it was rational in using time series data, where we were able to contrast historical data against current data, and to check whether there were any trends within our data inflating our results.

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